

Building a Reproducible Scholarly Search Toolkit

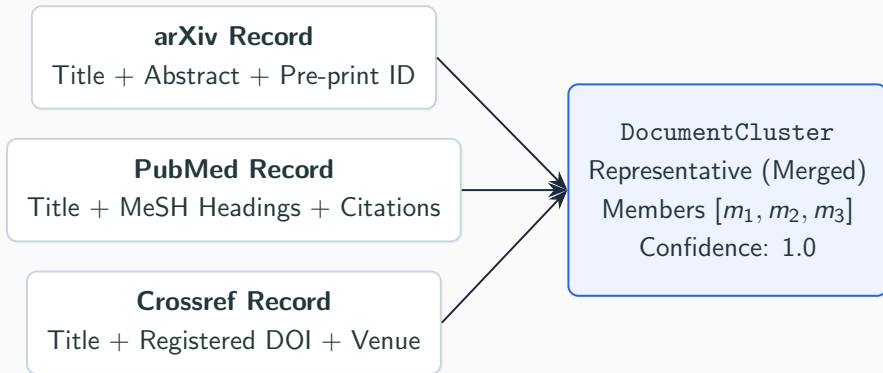
Episode 6: Non-Destructive Clustering

Nexus Scholar Suite

Non-Destructive Deduplication

Preventing Evidence Destruction

Naive deduplication permanently deletes duplicate records. We preserve every merged record inside a `DocumentCluster` while synthesizing an enriched representative.



The DocumentCluster Dataclass

```
@dataclass
class DocumentCluster:
    cluster_id: int
    representative: Document
    members: list[Document]

    @property
    def size(self) -> int:
        return len(self.members)

    @property
    def confidence(self) -> float:
        has_id = any(
            m.external_ids.doi or m.external_ids.arxiv_id
            for m in self.members
        )
        return 1.0 if has_id else 0.95
```